End-Term Sample Question Paper - SET B

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set B - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.

Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

Q1. Define an equivalence relation, name its three defining properties, and state how many equivalence classes congruence modulo 5 produces on $\mathbb{Z}$. **(2 Marks)**

[Unit I · CO1 · L1 - Remember · Theory]

Q2. For A = [[6, 3], [4, 5]], compute det A and state whether A is invertible, with the reason. **(2 Marks)**

[Unit II · CO2 · L2 - Understand · Numerical]

Q3. State the mean and the variance of a binomial random variable with n = 25 and p = 0.4. **(2 Marks)**

[Unit III · CO3 · L2 - Understand · Numerical]

Q4. State the chromatic number of the cycle C_6, the cycle C_7 and the complete graph K_5, with a one-line reason for the last. **(2 Marks)**

[Unit IV · CO4 · L1 - Remember · Theory]

Q5. State the three cases of the Master theorem for T(n) = aT(n/b) + f(n), and give the critical exponent that the cases compare f(n) against. **(2 Marks)**

[Unit V · CO5 · L2 - Understand · Theory]

Part B - Short Answer Questions (4 x 5 = 20 Marks)

Q6. Build the Huffman code for the symbol frequencies A = 2, B = 3, C = 4, D = 5, E = 6. List the merges in order and compute the total number of encoded bits. **(5 Marks)**

[Unit VI · CO6 · L3 - Apply · Numerical]

Q7. Find the multiplicative inverse of 9 modulo 31, and use it to solve $9x \equiv 5 \pmod{31}$ for the least non-negative x. State why the inverse exists. **(5 Marks)**

[Unit I · CO1 · L3 - Apply · Numerical]

Q8. Distinguish a discrete from a continuous random variable, define the expectation and the variance of a discrete random variable, and state what the variance measures. **(5 Marks)**

[Unit III · CO3 · L2 - Understand · Theory]

Q9. Define the classes P, NP, NP-hard and NP-complete, and explain in two sentences what a polynomial-time reduction is used for. **(5 Marks)**

[Unit V · CO5 · L4 - Analyse · Theory]

Part C - Long Answer Questions (3 x 10 = 30 Marks)

Q10. Eigenvalues and eigenvectors. **(10 Marks)**

[Unit II · CO2 · L4 - Analyse · Theory + Numerical]

(a) Define an eigenvalue and an eigenvector, state the characteristic equation, and state what the spectral theorem guarantees for a real symmetric matrix. *[5]*

(b) For A = [[7, 2], [2, 4]], find both eigenvalues and one eigenvector for each, and verify your eigenvalues using the trace and the determinant. *[5]*

Q11. Minimum spanning trees. **(10 Marks)**

[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]

(a) State what a minimum spanning tree is, how many edges it has on n vertices, and describe how Kruskal's algorithm builds one. *[5]*

(b) Apply Kruskal to the undirected graph with AB = 4, AC = 1, BC = 2, BD = 5, CD = 8, DE = 3, CE = 9. List every edge accepted and every edge rejected, and give the total weight. *[5]*

Q12. Dynamic programming for the 0/1 knapsack. **(10 Marks)**

[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]

(a) State the DP recurrence for V[i, w], and explain which two properties a problem must have before dynamic programming is worth using. *[5]*

(b) Build the complete table for weights {2, 3, 4}, values {25, 30, 50} and capacity W = 6, state the optimal value, and trace back the chosen items. *[5]*

----- End of Question Paper -----